

3: The circular motion of a model mounted on the merry-go-round.

Fig. 3.1 shows a bicycle wheel with the model mounted at a point at the end of a protruding arm, henceforth referred to as *the model point*. A push makes this model point move swiftly along the circular s -axis, and by use of the photo gate, we can follow how the velocity gradually slows down. Denoting the velocity in n -th passage by u_n , we let a computer plot a series of decreasing values $u_0, u_1, u_2, \dots$ as shown in fig. 3.2.

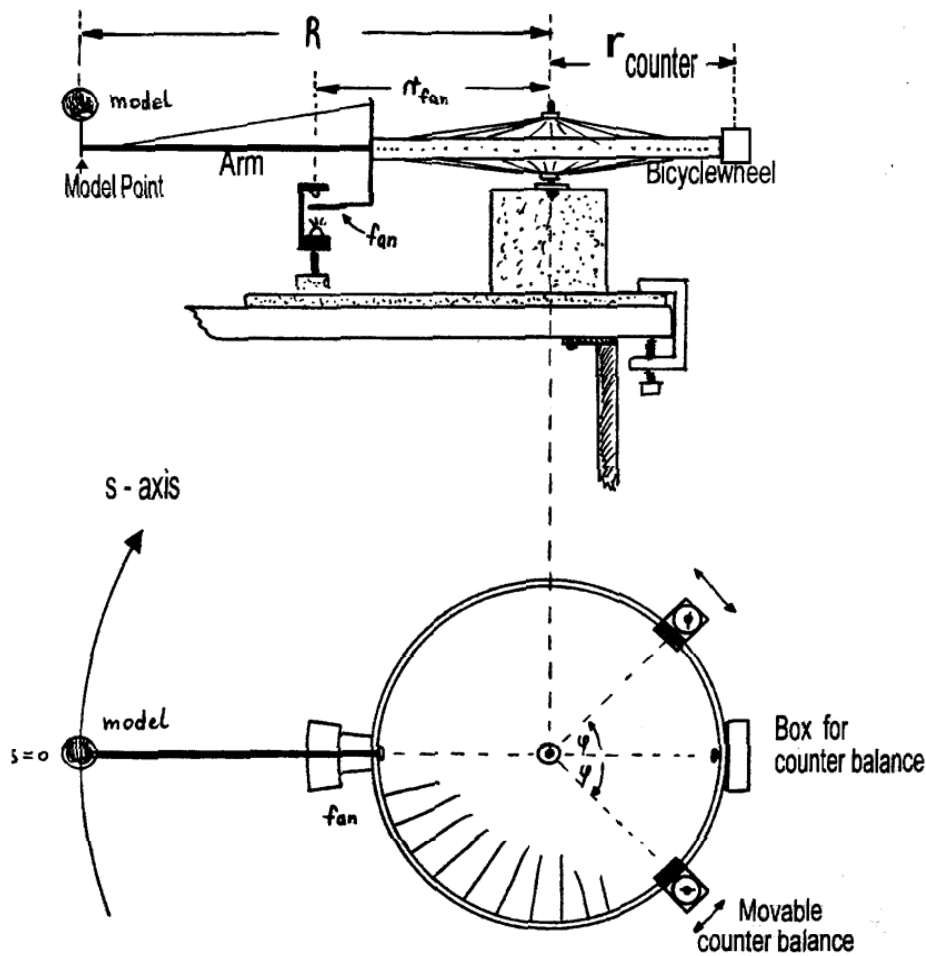


Figure 3.1: The merry-go-round carrying the model - in this case a smoothly surfaced sphere . With the counterbalances so adjusted (see subsection in sec. 5) that the centre of mass coincides with the axis of rotation, the merry-go-round is set in a quick rotation (by use of a counterbalance as a lever). The velocity u_n after n turns is determined by the time t_n it takes the fan to pass the photo gate

See also the photo in fig. 9.2, showing some extra details.

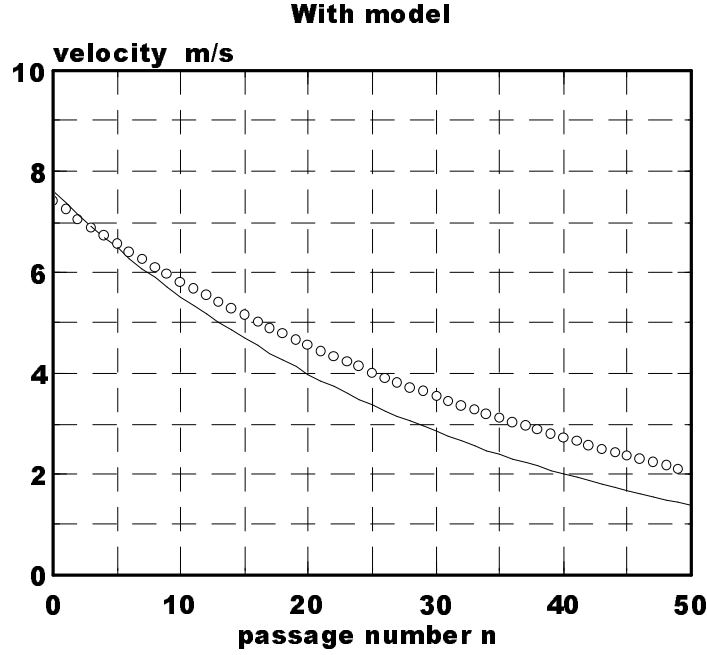


Figure 3.2: The small circles indicate the velocities $u_0, u_1, u_2, \dots$, measured with a smooth sphere (diameter = 62.5 mm) as the model. The solid line indicates the values of $v_0, v_1, v_2, \dots$ obtained with a certain guess for the three parameters v_0, α and β in eqn. (3.2). One sees directly that v_0 is chosen too large, and the values of α and β are chosen so that the speed goes down too quickly.

The decreasing experimental velocities can be reproduced very accurately by assuming the acceleration to be of the form

$$a = -\alpha v - \beta v^2, \quad (3.1)$$

where α and β are constants and v is the velocity. This assertion is verified as follows. Solving the equation of motion (3.1) (see appendix A), it is found that the velocity after n revolutions is given by

$$v_n = (v_0 + w) e^{-2\pi R \beta \cdot n} - w ; w = \frac{\alpha}{\beta}, \quad (3.2)$$

That is, the v_n decrease exponentially towards the negative limit $-w$. To check these theoretically expected velocities against the measured $u_0, u_1, u_2, \dots$, we make a first guess for v_0, α and β and let the computer draw the graph of the corresponding v_n (see fig. 3.2). A series of gradually improved guesses is made, and for each of these, the computer calculates *the least square sum*

$$\varepsilon = \sum_{n=0}^N (v_n - u_n)^2. \quad (3.3)$$

The smaller that is, the better is the guess. A "fairly good" guess¹ is easily ob-

¹ Actually it is quite simple to calculate a very good guess. The spreadsheet in fig. C2 does so by use of the method described in appendix D.

n	u_n	v_n	$v_n - u_n$
	m/s	m/s	mm/s
0	7.398	7.398	0.0
1	7.230	7.223	-7.0
2	7.044	7.052	8.3
3	6.888	6.885	-3.2
4	6.720	6.721	1.4
5	6.561	6.561	0.6
6	6.407	6.405	-2.2
7	6.248	6.252	4.4
8	6.100	6.102	2.2
9	5.954	5.956	2.3
10	5.810	5.813	2.8
11	5.670	5.673	2.6
12	5.533	5.536	2.6
13	5.399	5.402	2.6
14	5.269	5.271	1.7
15	5.141	5.143	2.0
16	5.014	5.017	3.8
17	4.890	4.895	4.4
18	4.774	4.775	1.3
19	4.655	4.658	2.5
20	4.541	4.543	1.5
21	4.428	4.431	2.2
22	4.321	4.321	-0.5
23	4.214	4.214	0.0
24	4.115	4.109	-6.8
25	4.010	4.006	-4.3
26	3.911	3.906	-5.2
27	3.814	3.807	-6.6

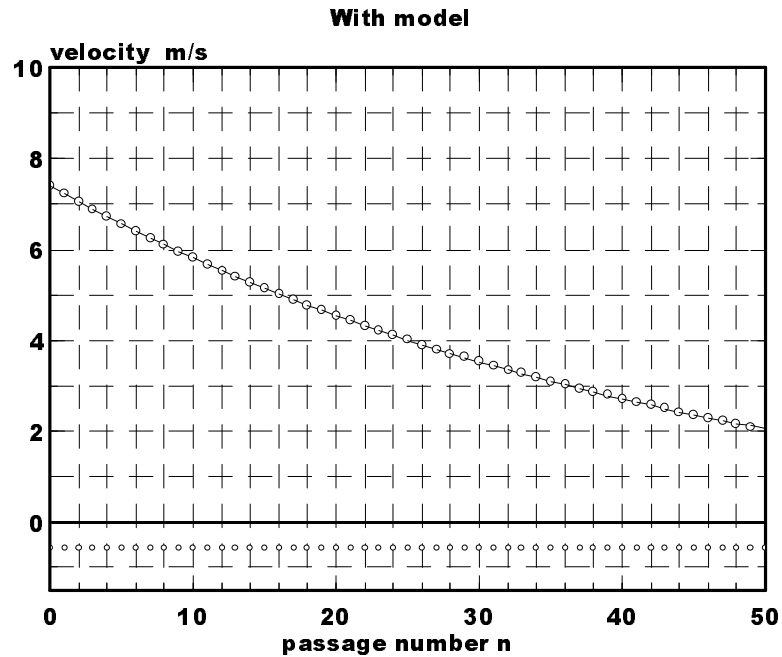


Figure 3.3: The same as in fig. 3.2 except that the solid curve now represents the best fit. The table to the left shows how small the first 27 differences are. Further down a few differences reach -10mm/s and these are the reason that the standard deviation become as high as $5,1\text{ mm/s}$ - still a rather small value.

The theoretical solid curve falls off exponentially towards the horizontal line of small circles at $v = -w$ underneath the horizontal axis.

tained and starting from that, the computer can determine the *best* guess i.e. the one which minimises ε . The corresponding values of v_0 , α and β are said to have been determined by *least square fitting*. The fit shown in fig. 3.3 is obtained in this way, and one sees that it so good, that it must be safe to conclude, that the model points acceleration within a *very* good approximation is given by the quadratic expression (3.2).

The slowing down of the model point seen in fig 3.3 is caused partly by air resistance against the model, partly by air resistance against the merry-go-round and partly by friction in the bearing. To be able to isolate the first of these causes, we repeat the experiment *without* the model and measure the corresponding values of the coefficients α and β . Again we see that the assumption $a = -\alpha v - \beta v^2$ is *extraordinarily* well founded. The slightly irregular values of the first few measured velocities illustrate something which happens in most runs. Almost certainly the relatively large deviations are caused by small "wobbles" in the arm to the model point, due to the abrupt start of the merry-go-round - wobbles that die away after a few revolutions. According to this explanation, one can just leave them out.

n	u_n	v_n	$v_n - u_n$
	m/s	m/s	mm/s
0	9.098	9.102	4.1
1	8.973	8.958	-15.3
2	8.809	8.815	6.4
3	8.665	8.675	9.9
4	8.539	8.536	-2.9
5	8.404	8.400	-4.0
6	8.263	8.265	2.3
7	8.133	8.133	-0.4
8	8.003	8.002	-0.7
9	7.876	7.873	-3.3
10	7.748	7.746	-2.3
11	7.619	7.620	1.8
12	7.497	7.497	-0.5
13	7.373	7.375	1.5
14	7.253	7.255	1.2
15	7.136	7.136	0.0
16	7.019	7.020	0.3
17	6.904	6.904	0.6
18	6.790	6.791	1.0
19	6.678	6.679	1.4
20	6.566	6.569	2.8
21	6.458	6.460	2.1
22	6.353	6.353	-0.5
23	6.248	6.247	-0.4
24	6.142	6.143	0.4
25	6.041	6.040	-0.8
26	5.940	5.939	-1.4
27	5.839	5.839	0.2

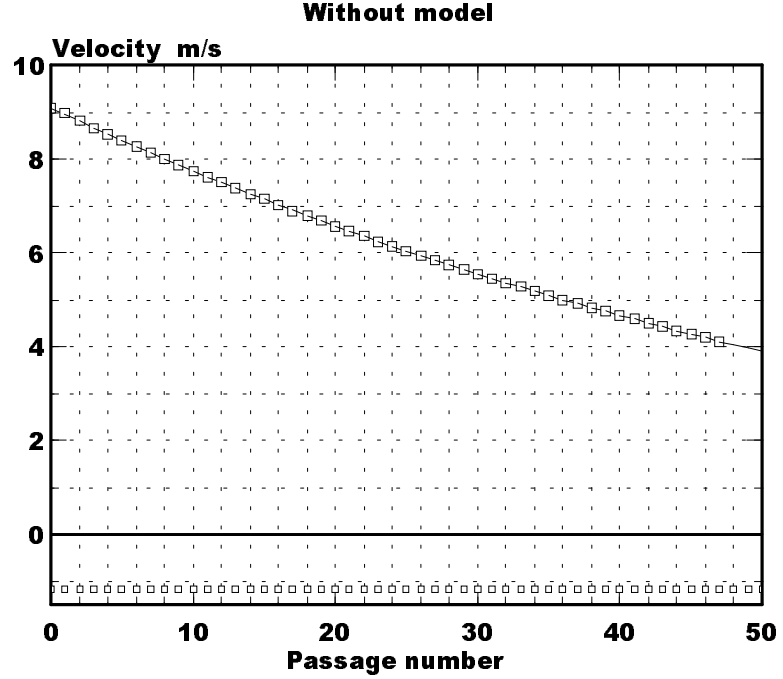


Figure 3.4: The same as in fig. 3.2, but for a run without the model fitted. Again the table to the left shows how small the first 27 differences are. This time the standard deviation for all the 47 measured velocities is only 3.2 mm/s. And most of this is due to the first four measurements. **Leaving these out, the standard deviation drops down to 1.3 mm/s.** An explanation for the slightly irregular start is given in the text.

The values of α and β determined for the runs shown in figs. 3.3 and 3.4 are collected in table 4.1.

A note on the accuracy of the best values of α and β

An interesting observation goes with the analytical expression (3.2) for the expected values of the velocities v_n . Assume that the series of decreasing velocities should come out as $u_0, u_1, u_2, \dots$, but for some reason come out as $\kappa u_0, \kappa u_1, \kappa u_2, \dots$ where κ is a certain common factor. This would happen if we have got the length r_{fan} in fig. 3.1 wrong - or the length of the fan which passes the photo gate. Or if the clock is wrong by a certain factor. It easily seen from the expression for v_n that if $u_0, u_1, u_2, \dots$ give v_0, α, β as the best values, then $\kappa u_0, \kappa u_1, \kappa u_2, \dots$ give $\kappa v_0, \kappa \alpha, \beta$ - the value α is multiplied by the same factor as the u_n , while the value of β remains unaffected. This, in turn, means that the function $C_d(v)$ which we are aiming at (see (4.6)) would come out as

$$C_d = \frac{\kappa c_1}{v} + c_2 \quad \text{instead of} \quad C_d = \frac{c_1}{v} + c_2 \quad (3.4)$$

Thus the value of c_2 is completely unaffected. This observation should be recalled when, in sec. 7, we identify C_d for car models with just c_2 .